

MATH1231 CALCULUS

Chapter 3C

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(3.8) Second Order Linear ODEs with Constant Coefficients

Linear Homogeneous

$$\frac{d^2y}{dx^2} + a\frac{dy}{dx} + by = 0 \quad (1)$$

Linear Non-Homogeneous

$$\frac{d^2y}{dx^2} + a\frac{dy}{dx} + by = f(x) \quad (2)$$

General Results

$$\frac{d^2 y}{dx^2} + a \frac{dy}{dx} + by = 0 \quad (1)$$

$$\frac{d^2 y}{dx^2} + a \frac{dy}{dx} + by = f(x) \quad (2)$$

Lemma 1 if y_1 and y_2 are solns of (1) then so is $\alpha y_1 + \beta y_2$

Lemma 2 (1) has two linearly independent solutions and every solution is a linear combination of these solutions.

Lemma 3 the general solution of (2) is given by $y = y_H + y_P$ where y_H is the general solution of (1) and y_P is a particular solution of (2).

(3.8.1) Homogenous Case

$$y''(x) + ay'(x) + by(x) = 0$$

There are two linearly independent solutions.

Trial Soln

$$y = e^{\lambda x}$$

$$\Rightarrow \lambda^2 e^{\lambda x} + a\lambda e^{\lambda x} + be^{\lambda x} = e^{\lambda x}(\lambda^2 + a\lambda + b) = 0$$

Hence $y = e^{\lambda x}$ is a soln **if and only if** λ is a root of the
characteristic equation

$$\lambda^2 + a\lambda + b = 0$$

Solutions of the characteristic equations $\lambda^2 + a\lambda + b = 0$

Case (i) $\lambda_1 \neq \lambda_2$ both real and distinct

$$y_H = Ae^{\lambda_1 x} + Be^{\lambda_2 x}$$

Case (ii) $\lambda_1 = \lambda_2$ both real and equal

$$y_H = Ae^{\lambda x} + Bxe^{\lambda x} \quad \lambda = \lambda_1 = \lambda_2$$

Case (iii) $\lambda_1 = \bar{\lambda}_2$ complex conjugate pairs

$$y_H = Ce^{\lambda x} + De^{\bar{\lambda} x} \quad \lambda = \lambda_1 \in \mathbb{C} \quad C, D \in \mathbb{C}$$

$$y_H = e^{\alpha x} (A \cos \beta x + B \sin \beta x) \quad \lambda = \alpha + i\beta \quad \alpha, \beta \in \mathbb{R}$$

Example

$$y'' + y' - 12y = 0 \quad y(0) = 0, \quad y'(0) = 1$$

Example

$$y'' - 4y' + 13y = 0 \quad y(0) = 0, \quad y'(0) = 1$$

(3.8.2) Non-homogeneous Case

$$y''(x) + ay'(x) + by(x) = f(x)$$

General solution $y(x) = y_H(x) + y_P(x)$ where

$$y_H''(x) + ay_H'(x) + by_H(x) = 0$$

We now know how to find $y_H(x)$, how do we find $y_P(x)$?

- ▶ inspired guesswork – method of undetermined coefficients
- ▶ variation of parameters – see MATH2120, MATH3201

Method of undetermined coefficients

$$y''(x) + ay'(x) + by(x) = f(x)$$

If $f(x)$ is a sum of terms (in column 1) then $y_P(x)$ is a sum of corresponding terms (in column 2) for certain values of the constants in the polynomials $Q(x)$

$f(x)$	trial $y(x)$
$P(x)$ given polynomial deg n	$Q(x)$ arbitrary polynomial deg n
$P(x)e^{ax}$	$Q(x)e^{ax}$
$P(x)e^{ax} \sin bx$ or $P(x)e^{ax} \cos bx$	$e^{ax} (Q_1(x) \sin bx + Q_2(x) \cos bx)$

Substitute the trial solution into the equation to find the constants

If any term in the trial solution is a solution of the homogeneous equation then multiply it by x

If still a solution of the homogenous equation then multiply by x again etc.

Example

$$y'' - 5y' + 6y = 12e^{5x}$$

Example

$$y'' - 5y' + 6y = 12e^{2x}$$

Example

$$y'' + y = \cos x$$

Example

$$y'' + 4y' + 4y = e^x$$

Example

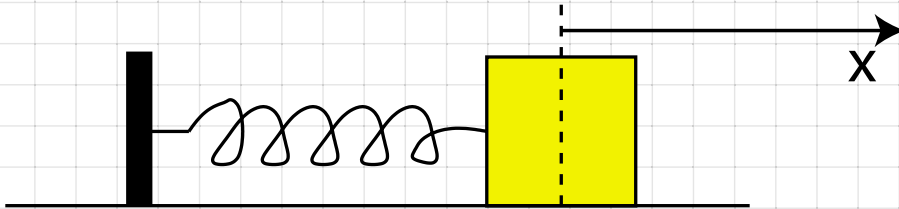
$$y'' + 4y' + 4y = e^{-2x}$$

Example

$$y'' - 4y' + 13y = x^2 \quad y(0) = 0, \quad y'(0) = 1$$

(3.83) Vibrations and Resonance

Consider a mass m at the end of spring on a frictionless table.



$$\frac{d^2x}{dt^2} + \omega^2 x = 0$$

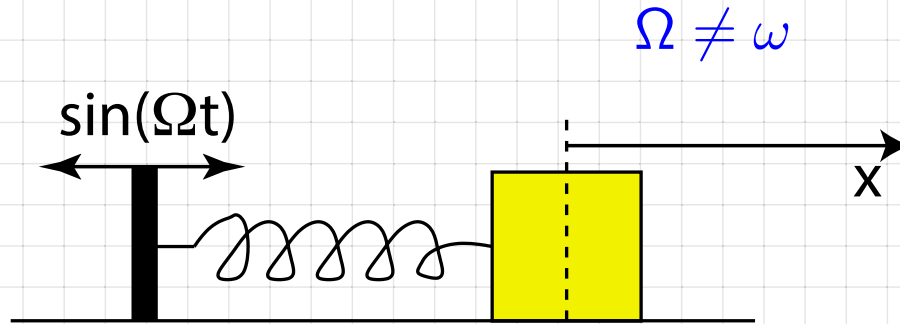
ω depends on the stiffness of spring

characteristic equation $\lambda^2 + \omega^2 = 0$

general solution $x(t) = A \cos \omega t + B \sin \omega t$

solutions are bounded and periodic

Vibrating Support



$$\frac{d^2x}{dt^2} + \omega^2 x = F \sin \Omega t$$

$$x_p = C \cos \Omega t + D \sin \Omega t$$

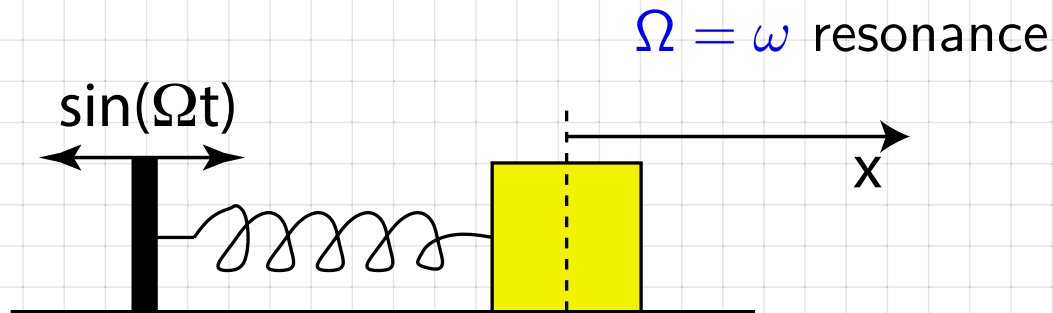
trial particular solution

$$x_p = \frac{F}{\omega^2 - \Omega^2} \sin \Omega t$$

general solution $x(t) = A \cos \omega t + B \sin \omega t + \frac{F}{\omega^2 - \Omega^2} \sin \Omega t$

solutions are bounded and oscillatory

Vibrating Support



$$\frac{d^2x}{dt^2} + \omega^2 x = F \sin \omega t$$

$$x_p = Ct \cos \omega t + Dt \sin \omega t$$

trial particular solution

$$x_p = -\left(\frac{F}{2\omega}\right) t \cos \omega t$$

general solution $x(t) = A \cos \omega t + B \sin \omega t - \left(\frac{F}{2\omega}\right) t \cos \omega t$

solutions are unbounded and oscillatory

Many mathematical models for real world behaviour reduce to

$$y''(t) + ay'(t) + by(t) = f_0 \sin \omega t$$

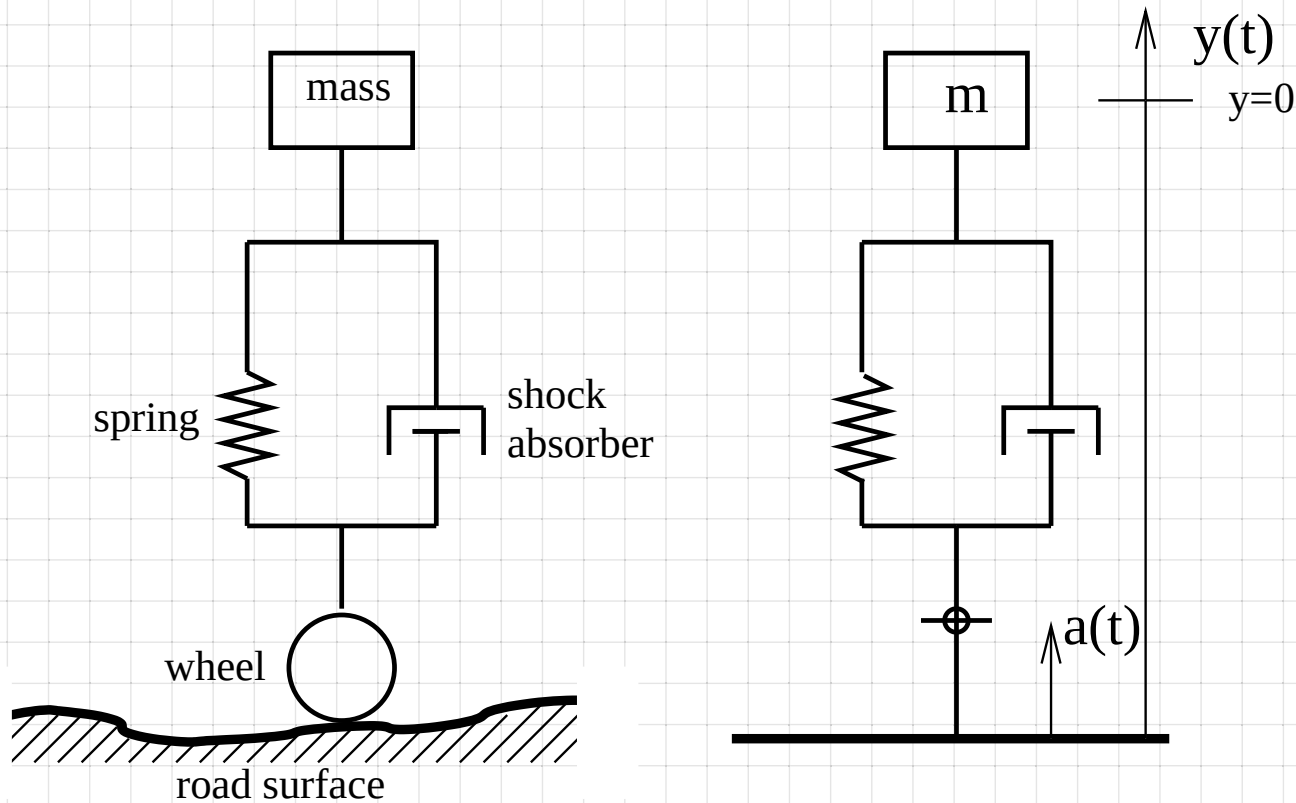
- ▶ The response $x(t)$ of an accelerometer to the rate of change in velocity in ground shaking $a(t)$ satisfies

$$x''(t) + 2\eta\omega x'(t) + \omega^2 x(t) = -a(t)$$

- ▶ The charge $q(t)$ on a capacitor in a series RLC circuit with an alternating field $f_0 \cos \omega t$ satisfies

$$Lq''(t) + Rq'(t) + \frac{q}{C} = f_0 \cos \omega t$$

Application – Car suspension and shock absorbers



In the reference frame of the car the road is moving up and down. This is $a(t)$

$$m \frac{d^2 y(t)}{dt^2} = -k(y(t) - a(t)) - c \frac{d}{dt}(y(t) - a(t))$$

$$my'' + cy' + ky = ca' + ka = f(t)$$

Smooth road, no forcing. No damping.

$f(t) = 0$ (no forcing) and $c = 0$ (no damping).

characteristic equation

$$m\lambda^2 + k = 0 \Rightarrow \lambda = \pm i\sqrt{\frac{k}{m}} \text{ then}$$

$$y(t) = A \sin \omega_0 t + B \cos \omega_0 t$$

where $\boxed{\omega_0 = \sqrt{\frac{k}{m}}}$ is the natural spring frequency

coil springs in a typical car have a natural frequency of approx 90 cycles per minute or $f = \frac{90 \text{ cycles}}{60 \text{ s}} = \frac{3}{2}$ Hz.

Angular frequency $\omega_0 = 2\pi f \approx 3\pi$ radians/s.

Smooth road, no forcing. With damping

$f(t) = 0$ (no forcing) but $c \neq 0$ (damping).

characteristic equation

$$m\lambda^2 + c\lambda + k = 0 \quad \Rightarrow \quad \lambda_{\pm} = \frac{-c \pm \sqrt{c^2 - 4mk}}{2m}$$

► **Overdamped** $c^2 > 4mk$

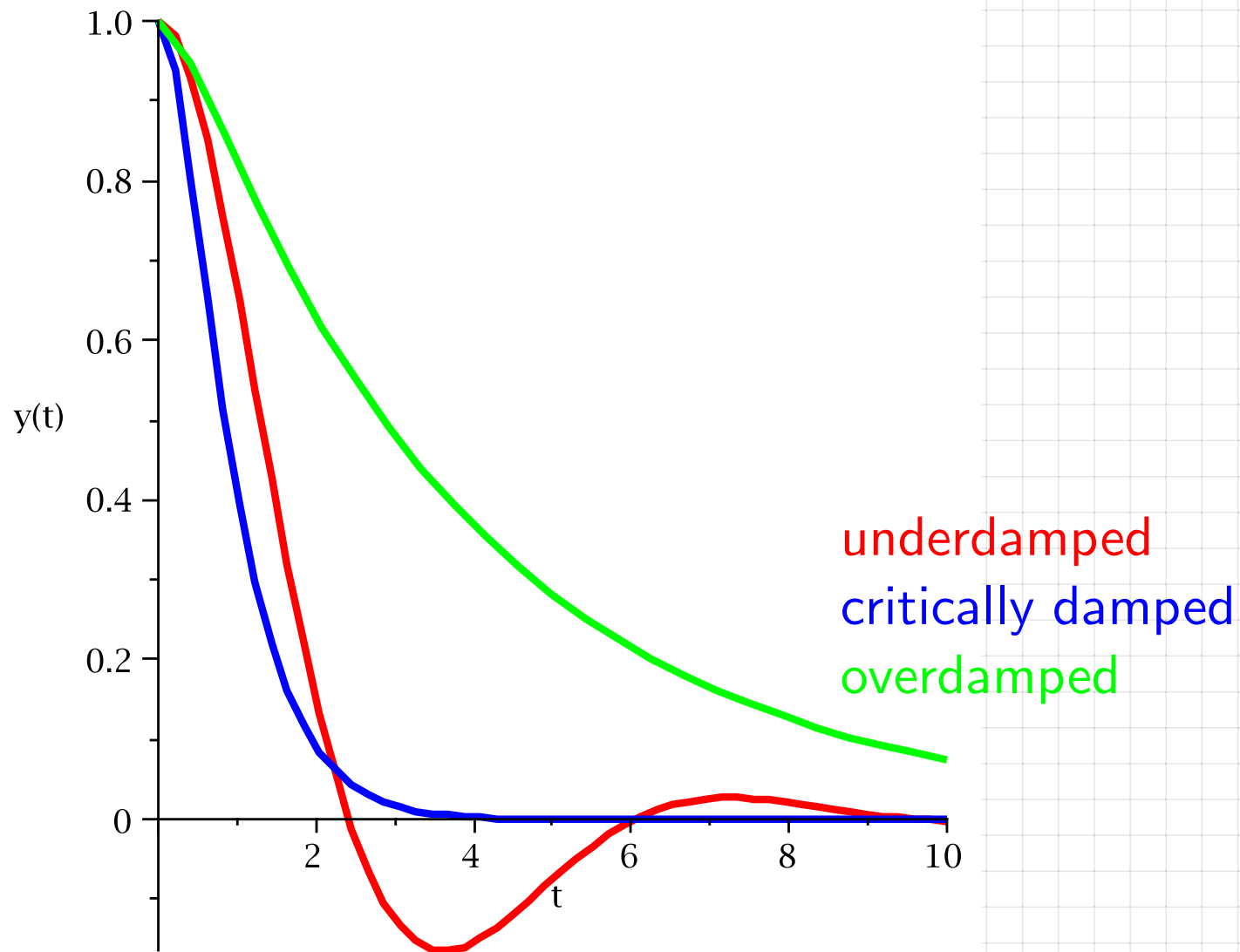
$$y(t) = Ae^{-|\lambda_+|t} + Be^{-|\lambda_-|t}$$

► **Underdamped** $c^2 < 4mk$

$$y(t) = e^{-\frac{c}{2m}t} (A \cos \Omega t + B \sin \Omega t), \quad \Omega = \sqrt{\omega_0^2 - \frac{c^2}{4m^2}}$$

► **Critically Damped** $c^2 = 4mk$

$$y(t) = Ae^{-\frac{c}{2m}t} + Bte^{-\frac{c}{2m}t}$$



Critical damping is the fastest way to return to equilibrium without oscillations (overshooting). Car shock absorbers are designed to achieve critical damping.

Bumpy road, gives forcing

Suppose the road surface has bumps on average ℓ units apart and each bump is a_0 units above the average level.

$$h(x) = a_0 \sin \frac{2\pi}{\ell} x$$



wheel motion approximation

Suppose that car is moving at constant speed v

$$a(t) = h(x = vt) = a_0 \sin \frac{2\pi v}{\ell} t$$

car suspension model

$$my'' + cy' + ky = F_0 \sin \omega t$$

Bumpy road, with forcing. With damping

$f(t) \neq 0$ (forcing) and $c \neq 0$ (damping).

- ▶ With forcing from a bumpy road we add in the **particular solution** $a \cos \omega t + b \sin \omega t$. (EX. Find a and b .)
- ▶ The amplitudes a and b are functions of the frequency ω . This defines a resonance frequency ω^* when a and b are maximal.
- ▶ $\omega^* = \sqrt{\frac{k}{m} - \frac{c^2}{2m^2}}$
- ▶ Regular road bumps can be spaced to create this resonance effect $\omega^* = \frac{2\pi v}{\ell}$.

